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Bi-Continuous W-Rich Refractory High Entropy Alloy-Cu Composite: Toward Material Innovation of Nuclear Reactor Coolant System

*Kook Noh Yoon, Il Hwan Kim, Ji Young Kim, Peter Hosemann, and Eun Soo Park**

Refractory high-entropy alloys (RHEAs) are considered promising candidate materials for next-generation nuclear reactors due to their superior mechanical strength, irradiation resistance, and thermal stability at high temperatures. However, the significant positive heat of mixing between refractory alloying elements and Cu, commonly used in cooling systems, poses challenges in forming composite structures. This study addresses the issue using a liquid metal dealloying (LMD) process. A precursor alloy (WTaVTi) with a directional dendrite-interdendrite structure is fabricated and reacted with molten Cu at 1200 °C for 96 h. This approach produced a RHEA-Cu composite with a stable interface between RHEA ($W_{31.5}Ta_{30.9}V_{21.4}Ti_{14.3}$) and Cu, featuring a spontaneously formed W-rich interlayer that enhances interfacial bonding. The composite showed excellent irradiation resistance, with 30% less swelling under α-ion irradiation than pure W. It also exhibited low thermal conductivity at room temperature, but reached ≈120 W $m^{-1}{\cdot}K^{-1}$ at ≈650 °C, surpassing pure W. This temperature-dependent rise in κ, with a positive gradient of +0.075 W $m^{-1}{\cdot}K^{-2}$, is attributed to decreasing diffuse mismatch at elevated temperatures. The large-scale reaction and stable microstructure achieved through LMD process highlight its industrial potential. This work offers a strategy for developing high-performance materials by combining RHEA's radiation resistance with Cu's thermal conductivity for extreme environments.

1. Introduction

Metal-matrix composites (MMCs) have garnered extensive research attention due to their unique combinations of properties that are unattainable with conventional materials. Among these, W-Cu composites stand out as one of the most successful MMC systems, widely utilized for both mechanical and functional applications.[1–3] These composites achieve enhanced strength and ductility by incorporating high strength but low toughness W particles, either as a dispersed second phase or interconnected ligaments, within a ductile but low strength Cu matrix, enabling their use in a variety of fields. Furthermore, the high thermal conductivities (κ) of Cu (386 W $m^{-1}{\cdot}K^{-1}$) and W (173 W $m^{-1}{\cdot}K^{-1}$) make these composites particularly suitable for applications requiring efficient heat dissipation, such as heat sinks leading to a combined thermal conductivity of as high as 238 W (m K)$^{-1}$ with 30% copper.[4] Owing to these properties, W-Cu composites are also employed in semiconductor packaging materials, where they facilitate rapid heat dissipation from electronic devices while maintaining mechanical durability. Additionally, they are used in high-voltage electrodes, which demand a combination of high strength and electrical conductivity.[5,6] Of particular note, these composites are considered promising candidates for use as joint materials in the diverters of fusion reactors.[7] In such applications, the bonding

K. N. Yoon, I. H. Kim, J. Y. Kim, E. S. Park
Department of Materials Science and Engineering
Research Institute of Advanced Materials & Institute of Engineering Research
Seoul National University
Seoul 08826, Republic of Korea
E-mail: espark@snu.ac.kr

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K. N. Yoon, P. Hosemann
Department of Nuclear Engineering
University of California
Berkeley
Berkeley, CA 94709, USA

P. Hosemann
Department of Materials Science and Engineering
University of California
Berkeley
Berkeley, CA 94709, USA

P. Hosemann
Department of Mechanical Engineering
University of California
Berkeley
Berkeley, CA 94709, USA

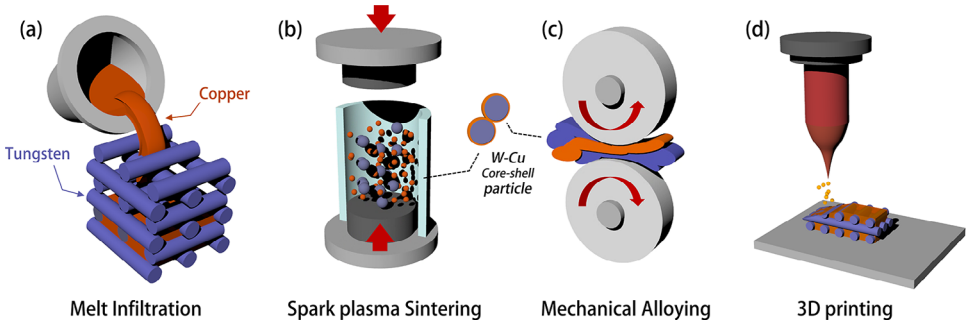


Figure 1. Schematic diagrams of various existing methods for fabricating W-Cu composites: a) Cu melt infiltration into a W skeleton, b) spark plasma sintering of W and Cu powders, c) mechanical alloying, and (d) 3D printing.

between plasma-facing materials (PFM) often envisioned as W based materials and Cu-based alloys (e.g., CuCrZr) used in cooling pipes requires not only high κ but also exceptional resistance to irradiation. Especially the low sputter yield, high melting temperature, and relatively low neutron activation make W and its alloys good plasm facing materials. Of course, enhancing thermal conductivity by adding Cu fosters an even better performance.

However, producing W-Cu composites with a homogeneous microstructure remains a significant challenge. Due to the large positive enthalpy of mixing (ΔH_{mix}: +22 kJ mol^{-1}) between W and Cu, these metals are not only difficult to alloy in the solid state but also tend to undergo phase separation even in the liquid state.[8] Moreover, the substantial difference in their melting and boiling points—W (≈3422 °C) has a melting point ≈1000 °C higher than the boiling point of Cu (≈2562 °C)—renders it impractical to create a uniform microstructure through conventional casting techniques such as induction melting or arc plasma melting.[9] (Further details on the separation behavior are provided in Figure S1, Supporting Information) Even when a composite structure is successfully formed, numerous interfacial defects, such as voids, are likely to develop between the separated phases, which can lead to reduced κ. Addressing the bonding issues between these phases is, therefore, a critical challenge.

To overcome these difficulties, various fabrication methods have been explored. One of the simplest approaches, as shown in **Figure** 1a, is to first prepare a W skeleton and then either pour Cu melt into the gaps or fill them with Cu powder and heat the assembly to form a composite structure.[10] However, since this method solidifies liquid Cu within solid W precursor, achieving a defect-free interface between the two phases can be challenging. Another approach involves using powders of both materials. As illustrated in Figure 1b,c, a composite containing coexisting phases can be created either by spark plasma sintering the W and Cu powders, or by mechanical alloying, applying severe plastic deformation.[11,12] While this approach can yield finer and more homogeneous microstructures, defects such as voids can still easily form at the interfaces. To address these issues, sintering core-shell structured alloy powders—produced by coating W particles with Cu vapor—have emerged as one of the most promising approaches for creating defect-free W-Cu composites, despite the complexity of the process.[13] This method ensures near-perfect contact between the phases, effectively minimizing heat loss during thermal transfer—a significant advantage for high-performance applications. Moreover, the use of metal 3D printing technologies has recently gained attention as another alternative for fabricating these composites.[14]

The Liquid Metal Dealloying (LMD) process, introduced by Wada et al., has emerged as a groundbreaking technique for fabricating MMCs that are challenging to produce using conventional methods.[15] In this process, a precursor binary alloy is exposed to the liquid phase of a third metal. The liquid dissolves one element (reactive with the liquid) from the precursor and infiltrates the binary alloy leaving a defect-free interface between the infiltrated element and one of the precursor elements (non-reactive).[16] Leveraging this method, McCue et al. successfully synthesized Ta-Cu composites by reacting Ta-Ti alloy precursors with Cu melts.[17] Building on this innovative approach, the present study employs the LMD process to fabricate composite materials comprising an irradiation-resistant W-rich refractory high-entropy alloy (RHEA) and Cu.[18,19]

In this study, WTaVTi RHEA, characterized by a microstructure of W-rich dendrites and Ti-rich interdendritic (ID) regions, was used as the precursor in the LMD process. Preferential reactions were induced in the ID regions within a pure Cu melt, ensuring robust thermal conductivity (κ) in the Cu-rich phase formed by the process. This approach facilitated faster and more uniform reactions, maximizing the synergistic properties of each phase. The resulting RHEA-Cu composite exhibited superior κ compared to pure W at temperatures exceeding ≈650 °C. Moreover, this composite retains the inherent HEA characteristic of increasing κ with an increase of temperature, thereby serving as an even more effective heat transfer material at elevated temperatures. Consequently, owing to this attribute, the composite holds significant promise for high-temperature applications—such as in nuclear reactor environments—where it not only provides the enhanced mechanical properties and irradiation resistance afforded by its composite structure, but also delivers superior thermal conductivity. To further investigate the mechanisms underlying these remarkable properties, detailed microstructural and phase characterizations of each phase were conducted using TEM analysis. The RHEA-Cu composite developed in this study holds significant promise for diverse applications, not only as a foundational material for next-generation nuclear reactors and fusion power plants but also for thermal dissipation in extreme operational environments.

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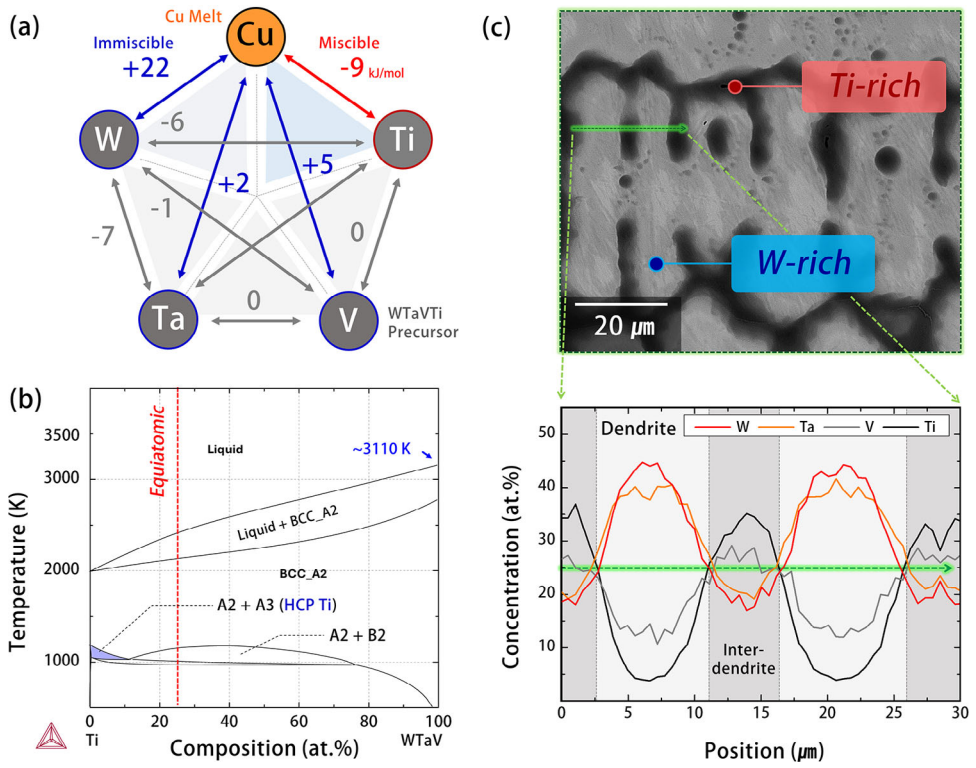


Figure 2. Thermodynamic design of the precursor alloy for the LMD process. a) Thermodynamic relationships between the Cu melt involved in the LMD process and the alloying elements with positive and negative ΔH_{mix} with Cu. b) Pseudo-binary phase diagram of the precursor material, WTaVTi alloy. The diagram is constructed with WTaV, which exhibits positive ΔH_{mix} with Cu, as one axis based on thermodynamic considerations. c) Microstructure of the precursor with D-ID structure and its EDS results.

2. Results

2.1. Alloy Design of WTaVTi Precursor for the LMD Process

To ensure the success of the LMD process, careful design of the precursor alloy is essential to optimize its interaction with the metallic melt. During the LMD process, selective atomic exchange, surface diffusion, and spontaneous atomic passivation play critical roles. The precursor alloy must therefore contain elements with both positive and negative ΔH_{mix} with the melt (Cu). Through this mechanism, elements with negative ΔH_{mix} dissolve into the melt, leaving behind a composite structure composed of elements with positive ΔH_{mix} and the melt.[16] In this study, the quaternary WTaVTi alloy, recognized for its excellent mechanical properties, was chosen as the precursor material. This alloy was reacted in molten Cu bath to form a composite structure comprising RHEA and Cu.[19]

Figure 2a depicts the thermodynamic relationships between the elements in the WTaVTi RHEA and the Cu. This figure highlights why Ti is the most suitable sacrificial element for the LMD reaction employed in this study. An ideal sacrificial element for the LMD process should satisfy the following criteria: 1) it should readily form a solid solution with other alloying elements in the precursor alloy without forming any secondary phases. 2) it should exhibit a negative ΔH_{mix} with the liquid metal (Cu), and 3) it should possess a relatively low melting point to enable rapid diffusion in the molten metal bath. Ti meets all of these requirements. Specifically, Ti is known to form a disordered BCC (A2) solid solution with W, Ta, and V, ensuring stable precursor alloy formation. At the same time, it exhibits a negative ΔH_{mix} of –9 kJ mol^{-1} with Cu, in stark contrast to the other RHEA constituents—W, Ta, and V—which exhibit positive ΔH_{mix} values of +22, +2, and +5 kJ mol^{-1}, respectively.[8] Furthermore, Ti has the lowest melting point (1668 °C) among Group 4 elements, enabling faster diffusion in the melt. Consequently, Ti is preferentially selected as the sacrificial element that undergoes atomic exchange with Cu and is subsequently removed from the precursor, while Group 5 and 6 transition metals, including W, remain in the HEA phase due to their immiscibility with Cu.

Figure 2b presents the pseudo-binary phase diagram of the WTaVTi alloy tailored for the LMD process. Using Ti as the sacrificial component on one axis and assuming complete solid solution formation for WTaV on the other, the diagram predicts a uniform BCC solid solution phase at temperatures above 1200 K (≈930 °C). The liquidus temperature of the WTaV phase, which constitutes the ligaments in the RHEA-Cu composite, is ≈3110 K, indicating the possibility of having excellent high-temperature mechanical properties. Based on this phase diagram, a precursor alloy with an equiatomic composition of all four elements was selected. This composition includes ≈25 at.% Ti, which is expected to yield a composite with an estimated 30 vol% of Cu-rich phase, if all of the Ti atoms had position exchange with Cu.

Besides, formation of the ordered B2 phase is also predicted in compositions containing 15–75 at.% Ti at temperatures below

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Table 1. Compositional constituent of as-cast precursor, and RHEA-Cu composite fabricated by LMD process.

Alloy composition		Contents [at.%]					Phase fraction [%]	
		W	Ta	V	Ti	Cu	Volume	Weight
As-cast RHEA (Precursor)	Overall	25.38	26.57	25.08	22.97	–	100.00	–
	Dendrite	37.46	28.91	17.82	15.81	–	71.15	–
	Inter-dendrite	9.85	17.7	35.84	36.61	–	28.85	–
After LMD process (Composite)	Dendrite	31.45	30.88	21.35	14.33	1.99	63.61	71.82
	Inter-dendrite	0.27	0.35	0.14	0.09	99.15	36.39	28.18

1150 K (≈0.48Tm). Nevertheless, the formation of this phase is unlikely due to the rapid quenching employed during the precursor casting process using water-cooled Cu molds. Furthermore, in Ti-rich compositions with more than 85 at.% Ti, the formation of the disordered HCP (A3) phase is expected in similar temperature ranges (indicated in the blue region of the diagram). However, in the equiatomic precursor composition utilized in this study, which contains over 25 at.% Ti, the formation of the HCP phase during conventional alloy casting is improbable.

Figure 2c illustrates the microstructure of the as-cast precursor material, emphasizing the RHEA dendrites that will form the ligament regions in the composite. The lower panel of the figure presents the compositional distribution of the dendritic-interdendritic (D-ID) structure, analyzed using EDS. According to the EDS line scan, the dendrites are enriched with high-melting-point elements W and Ta, exhibiting a composition of $W_{37.5}Ta_{28.9}V_{17.8}Ti_{15.8}$. In contrast, the ID regions are enriched with low-melting-point elements Ti and V, with a composition of $W_{9.9}Ta_{17.7}V_{35.8}Ti_{36.6}$. Detailed data provided in **Table 1** indicate a D-ID phase volume ratio of ≈7:3. This segregation occurs because, during solidification, the high-melting-point elements W and Ta preferentially form dendrites, while the low-melting-point elements Ti and V are redistributed into the liquid phase, forming the ID regions.

Additionally, Figure 2c reveals droplets within the dendrites, ranging in size from several tens of nanometers to ≈3 μm in diameter, which exhibit a contrast similar to that of the ID regions. These droplets are microstructural features formed by capillary-driven instabilities during the coarsening of dendrites and the solidification of liquid remnants.[20] Droplet formation is more likely during solidification when a material exhibits a wide liquid-solid coexistence temperature range. In the WTaVTi alloy, this temperature gap is ≈400 K during dendritic solidification, making it particularly susceptible to droplet formation. Nevertheless, since their volume fraction is relatively small, they are not expected to significantly influence the properties of the final composite. Additionally, these droplets possess almost identical chemical compositions to the ID regions since they are formed during solidification. Therefore, if exposed on the surface, they are likely to participate in the LMD reaction, resulting in Cu occupying these positions in the final composite.

Previous studies on the LMD process have predominantly utilized precursors with homogeneous atomic structures, such as single-phase metals or amorphous materials.[15] These homogeneous materials allow for uniform LMD reactions and provide easier control over the final microstructure. However, the resulting Cu ligaments typically form isotropic, interconnected networks throughout the material. In contrast, the naturally solidified WTaVTi alloy introduces connected Ti-rich ID channels within the precursor material. These ID regions serve as preferential reaction sites during interactions with Cu since they contain relatively higher Ti contents than dendrites. Furthermore, the alignment of these channels along the solidification direction is advantageous for establishing thermally conductive pathways within the composite.

2.2. WTaVTi Precursor's Mechanical Property and Irradiation Resistance

RHEAs are renowned for their exceptional mechanical properties at both ambient and elevated temperatures.[21,22] These attributes, particularly their high-temperature strength and phase stability, are critical for their application in nuclear materials. To predict the mechanical properties of the final composite produced via the LMD process, the compressive stress-strain curves of as-cast RHEAs, including the WTaVTi precursor alloy, were measured, as shown in **Figure 3a**. To investigate the effect of alloying elements on mechanical properties, stress-strain curves were analyzed for a series of as-cast RHEAs, beginning with unary pure W and sequentially adding elements to form binary (WTa), ternary (WTaV), and quaternary (WTaVTi, the precursor) systems.

As shown in the figure, pure W exhibited a maximum elongation of ≈44.9% and a yield strength (YS) of ≈563 MPa. In contrast, the binary W-Ta alloy showed more than double the YS (≈1159 MPa) but significantly reduced ductility (≈12.5%). With the addition of V and Ti to the W-Ta binary alloy, the maximum elongation increased to 15.7% and 19.4%, respectively. This trend is consistent with previous studies showing that incorporating elements with fewer d-shell electrons, such as Group 4 elements, into RHEAs enhances their plasticity.[23] Furthermore, the addition of V and Ti gradually increased the YS to 1267 MPa and 1467 MPa, respectively, due to the solid-solution strengthening effect of alloying elements with differing atomic sizes. Notably, the ternary WTaV alloy, expected to form the ligament structure in the RHEA-Cu composite, exhibited a slightly lower YS (≈200 MPa less) than the quaternary alloy but achieved a higher ultimate tensile strength (UTS) of ≈1950 MPa. Additionally, RHEAs with excellent room-temperature properties retained high YS values at elevated temperatures, a key requirement for the current

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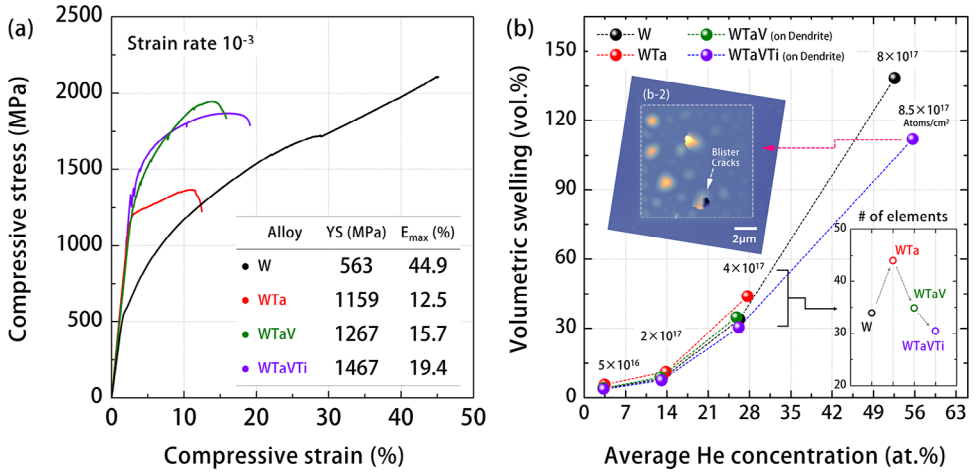


Figure 3. As-cast WTaVTi alloy with excellent mechanical properties and irradiation resistance. a) Compressive stress-strain curves of four alloys: W, WTa, WTaV, and WTaVTi. b) He-induced volumetric swelling of each alloy as a function of He concentration under α-ion (He+) irradiation. (b-2) AFM image showing surface blister cracking caused by alpha-ion irradiation.

molten salt reactors operating at ≈700 °C. These alloys maintained YS values above 850 MPa at 700 °C and over 800 MPa at extreme temperatures exceeding 1000 °C, where superalloys such as Inconel typically undergo significant degradation. Even at ultra-high temperatures above 1600 °C, the alloys sustained strengths exceeding 400 MPa, highlighting their superior high-temperature performance.[24]

Furthermore, studies by Kim et al. demonstrated that the quaternary W-Ta-V-Ti alloy exhibits outstanding properties even at elevated temperatures of 800 °C.[18] For instance, while pure W exhibited a high room-temperature YS of 641 MPa, its strength dropped precipitously with increasing temperature, reaching only ≈100 MPa at 400 °C and remaining similarly low at 800 °C. In contrast, the W-Ta-V-Ti alloy exhibited a remarkable room-temperature YS of over 1400 MPa. Although it also softened at elevated temperatures, it maintained a YS of 842 MPa at 400 °C and displayed hardening anomalies at higher temperatures, achieving 978 MPa at 600 °C and 863 MPa at 800 °C. (Further details, including stress-strain curves, are provided in Figure S2, Supporting Information). These exceptional high-temperature properties suggest that the composite fabricated from this high strength WTaVTi alloy could be effectively utilized at even higher operational temperatures.

Although the full strength of the alloy may not be retained in the composite, the strength of a composite is linearly correlated with the strength of its constituent materials.[25] For instance, the RHEA-Cu composite features a structure where Cu ligaments are interconnected within the RHEA matrix, resembling a directional filler system. Assuming a stable interface between the two phases, the composite strength can be estimated based on the volume fraction of each phase.[26a]

At this point, the two most representative methods for estimating the strength of a composite are as follows. Under Voigt's assumption of isostrain, where the entire material deforms uniformly under the same strain, the composite strength can be expressed as: $\sigma = \sigma_m V_m + \sigma_f V_f$. In contrast, under Reuss's assumption of isostress, where the material experiences uniform stress but different strains, the composite strength is given by: $\frac{1}{\sigma} = \frac{V_m}{\sigma_m} + \frac{V_f}{\sigma_f}$. In practical cases, the YS of the composite generally falls between the predictions of Voigt's and Reuss's assumptions. The YS, defined as the stress at the end of the elastic region, can thus be approximated using these equations.

Meanwhile, MMCs fabricated via the LMD process tend to exhibit robust interfacial bonding, as the interfaces are formed through spontaneous atomic interdiffusion during the reaction. This significantly reduces the possibility of interfacial debonding and ensures mechanical continuity between phases. For example, in a Ti–Mg bicontinuous composite fabricated via LMD, the experimentally measured strength closely matched the value predicted by Voigt's upper bound approximation.[26b] This result highlights the fact that the interfaces formed through LMD are well bonded at the atomic scale. As will be further discussed below, a similar bonding is expected to form between the two phases of the RHEA–Cu composite developed in this study. Therefore, it is reasonable to expect that the excellent mechanical properties measured in the RHEA precursor can be largely retained in the composite structure.

Moreover, when the robust interfacial bonding is present, and the volume fraction of the stiffer phase is sufficiently high and interconnected well, the load during deformation tends to be dominantly supported by that phase rather than being limited by interfacial failure. In such cases, the overall strength of the composite is largely governed by the mechanical behavior of the stronger and stiffer phase. Given that the RHEA–Cu composite developed in this study consists of a relatively soft Cu matrix and a RHEA ligament with a high volume fraction (≈64 vol.%), it is reasonable to anticipate that the excellent strength of the RHEA precursor is effectively inherited by the final composite. This expectation is supported by previous work by Wenqi et al.,[26c] who investigated the strength evolution in W–Cu composites as a function of W content. When the W content was relatively low (60 wt.%, ≈40 vol.%), the softer Cu phase dominated the composite strength. However, when the W content increased to 80 wt.% (≈65 vol.%), forming an interconnected network of W, the yield strength of the composite was primarily governed by the W phase. This

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demonstrates that, once the stiffer phase exceeds a certain critical volume fraction and percolates through the structure, it becomes the dominant contributor to the composite's mechanical behavior. Accordingly, the RHEA–Cu composite developed in this study, with its continuous RHEA ligament occupying over 64 vol.%, is expected to retain the high mechanical strength of the precursor. A more detailed analysis of the composite's mechanical properties, however, is beyond the scope of the present study and will be addressed in future work.

As mentioned above, understanding the radiation response of the materials is a critical element for deploying materials in nuclear environments. W, the primary material selected for the RHEA in this study, is well studied in neutron as well as ion irradiation damage, and the leading candidate for fusion first wall components. It has been shown that HEAs demonstrate superior irradiation resistance compared to conventional dilute alloys.[27–29]

One possible reason for the enhanced radiation tolerance observed in HEAs is their intrinsically complex lattice structure. The significant chemical compositional fluctuations in HEAs promote high interstitial-vacancy recombination rates, enabling them to resist irradiation-induced damage more effectively. Additionally, the distorted lattice structure in HEAs slows atomic diffusion, thereby limiting defect formation. Notably, the distorted lattice does more than merely reduce diffusion: for instance, it may also diminish bias effects between vacancies and interstitial elements, in turn alleviating swelling and further mitigating radiation damage. However, as some have suggested, the heightened lattice strain in HEAs can reduce bias effects between vacancies and interstitials, resulting in less swelling and a corresponding improvement in radiation tolerance. In other words, the lower diffusion speed alone may not fully account for the enhanced irradiation resistance. On the other hand, some argue that the reduced mass diffusion caused by the distorted lattice structure of HEAs may suppress recombination, potentially weakening irradiation resistance. However, some recent studies have reported contrasting findings, suggesting that lattice distortion can, in fact, enhance recombination.[30,31] For instance, thermal spikes generate localized heat under ion-irradiation with Ni or Xe, and due to the intricate lattice structure of HEAs, this heat is not easily dissipated. As a result, the retained heat facilitates defect recombination rather than suppressing it. Although a similar effect might be expected in our study, where lighter He ions were used, further research is needed to validate this phenomenon.

To evaluate irradiation resistance using a fast-screening method, dendrites of the RHEAs were subjected to α-ion (He^+) irradiation, and the degree of swelling was measured as a function of Helium concentration.[32–34] This dendritic region will be changed to the high-strength, irradiation-resistant RHEA ligament by the LMD process. Therefore, evaluating the properties of this region serves as a reasonable proxy for inferring the overall behavior of the composite. While this material itself is not considered directly as plasma facing material yet even the transition material will be exposed to high energy neutron irradiation and therefore transmutation reactions generating large amounts of Helium are a concern. Helium is insoluble in materials and will form bubbles especially in combination with displacement damage and elected temperatures. These bubbles can lead to enhanced swelling as well as further embrittlement, especially at grain boundaries. Of course, while the mechanisms of helium introduction differ, both can result in comparable microstructural evolution. He ion irradiation provides a relevant and efficient means to simulate helium-induced effects such as swelling under fusion-relevant conditions. Consequently, screening for related effects is essential to highlight if the new material behaves comparable to already studied materials. This study does not intend to shed detailed light into the radiation response of this new class of materials here but provides a data point using a rapid throughput radiation effects scoping method like low energy localized helium radiation using a Helium Ion Beam (HIM) microscope as outlined in previous work.

In this study, 25 keV α-ions were implanted into the materials, and He induced swelling was analyzed using AFM. Figure 3b shows the swelling height of each material as a function of the average implanted He concentration. All alloys displayed increasing swelling height with the increase of the He concentration. (The dpa and He concentrations, which vary with the irradiation depth for each alloy, were calculated using SRIM simulation. The detailed results are presented in Figure S3, Supporting Information). For pure W and the WTaVTi precursor alloy, blistering was observed under sustained irradiation. Up to ≈27 at.% of He implantation (fluence ≈4×10^{17} atoms cm^{-2}, ≈15 dpa), all alloys experience similar behavior under this condition.

As the number of implanted ions increases, defect clustering leads to blistering, which eventually results in surface cracking. This cracking mechanism releases strain energy, halting further swelling. To validate this mechanism, α-ion implantation was conducted on pure W and the WTaVTi RHEA until cracks formed, as shown in Figure 3(b-2). As shown in this figure, most of the observed swelling is understood to result from the formation of helium bubbles. Both materials showed sharp increases in swelling height beyond a fluence of 4×10^{17} atoms cm^{-2} (≈14 at.% of He concentration), with cracks appearing at 8×10^{17} atoms cm^{-2} for W and 8.5×10^{17} atoms cm^{-2} (≈60 at.% of He concentration)for WTaVTi. The swelling heights at these fluences were 137% for W and 108% for WTaVTi. Notably, the RHEA exhibited ≈30% lower swelling and resisted blistering until a fluence ≈4×10^{17} atoms cm^{-2} higher than pure W. These findings underscore the potential of the WTaVTi alloy as an advanced material for nuclear and fusion cooling systems, combining reduced swelling and greater resistance to blistering-induced failure.

Swelling behavior varied slightly with alloy composition, as shown in the inset of Figure 3b. At a fluence of ≈4×10^{17} atoms cm^{-2}, pure W exhibited 32% swelling, which increased to 44% in the WTa alloy. However, the addition of V reduced swelling to levels similar to pure W, and the WTaVTi alloy demonstrated the lowest swelling (≈29.5%). Of course, the difference between W's 32% and 29.5% may be considered insignificant. Nevertheless, it is worth emphasizing that the newly developed WTaVTi alloy exhibits properties comparable to those of W. This excellent resistance might be attributed to the restricted dislocation migration caused by the distorted lattice structure of HEAs, making them highly promising materials for nuclear applications.[35]

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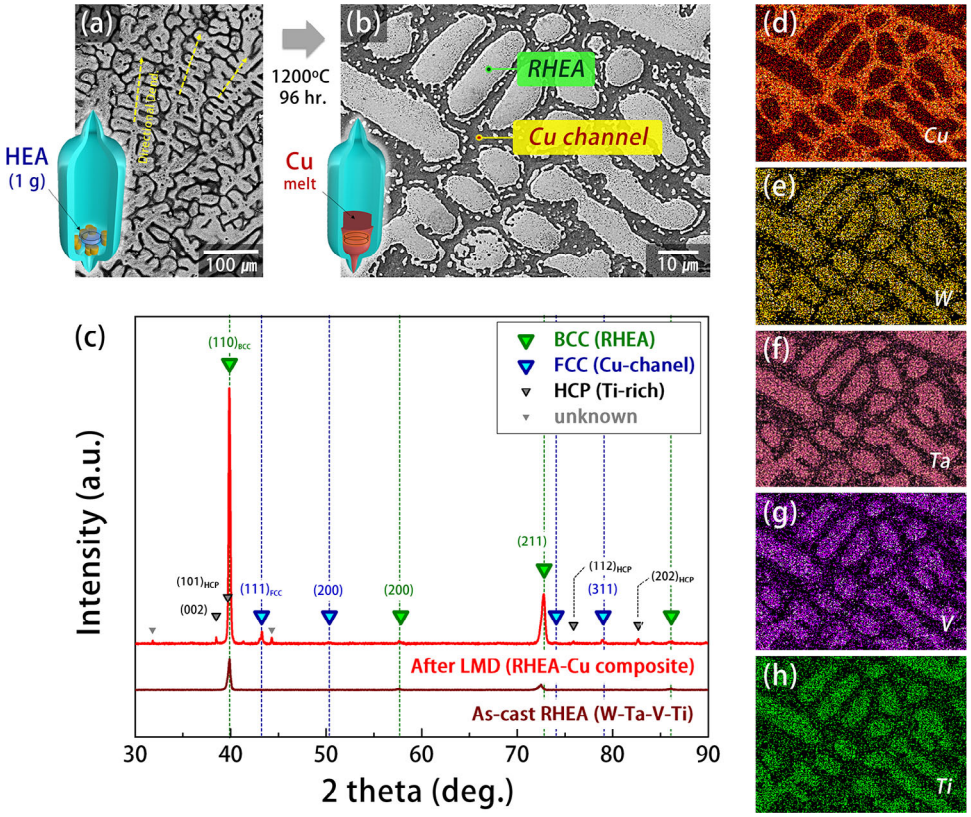


Figure 4. Microstructures before and after the LMD process. a) Microstructure of the as-cast WTaVTi precursor prior to the LMD process. b) Microstructure of the composite formed after reacting with a Cu melt at 1200 °C for 96 h, showing the complete substitution of the ID region in Figure 4a with Cu. c) XRD traces before and after the LMD reaction, along with EDS maps illustrating the elemental distribution post-reaction: d) Cu, e) W, f) Ta, g) V, and h) Ti.

2.3. Fabrication of RHEA-Cu Composite via LMD Process

A unique RHEA-Cu composite was successfully fabricated by reacting the WTaVTi RHEA in a Cu melt bath. **Figure** 4a,b illustrate the WTaVTi precursor alloy and the resulting composite microstructure obtained through the reaction. As shown in Figure 2 and Table 1, the precursor alloy exhibited a D-ID structure in the as-cast state, with the ID region containing ≈36.6% Ti, which is prone to dissolve into Cu. The average dendritic first-arm thickness was ≈11.3 μm (±1.42), accounting for ≈71% of the total volume fraction, while the Ti-rich ID regions displayed spacings of 2–5 μm depending on location. The XRD trace (brown) in Figure 4c confirmed that the as-cast RHEA consisted of a single BCC phase. With this single-phase precursor alloy, the RHEA precursor and Cu melt were sealed in a silica tube under vacuum. The precursor, shaped as a 1-mm-thick, 4-mm-diameter coin to comply with LFA ASTM standards, was reacted at 1200 °C for 96 h. According to the phase diagram in Figure 2b, this temperature ensured the RHEA maintained a complete solid solution without phase separation or precipitate formation, facilitating a homogeneous reaction.

The resulting composite microstructure, shown in Figure 4b, retained bright and dark contrasts, indicating the formation of distinct regions due to the interaction between Cu and Ti in the ID regions. The XRD results (red trace) in Figure 4c confirmed a composite structure comprising BCC RHEA ligaments and FCC Cu channels. A small fraction of HCP phases was also detected, aligning with predictions from the pseudo-binary phase diagram in Figure 2b, which suggested Ti-rich HCP precipitates formed during cooling. Minor unindexed peaks, likely attributable to intermetallic phases or oxides formed during the reaction, were observed but had minimal intensity, indicating negligible volume fractions and little impact on composite properties.

Figure 4d–h present EDS maps of Cu, W, Ta, V, and Ti, showing their distribution within the fabricated composite. After the reaction, the Ti-rich ID regions predominantly exchanged into Cu, with a purity exceeding 99%, while the dendritic regions retained the remaining alloying elements. The ID regions, initially composed of $W_{9.9}Ta_{17.7}V_{35.8}Ti_{36.6}$, transformed into pure Cu channels, while the dendrites, initially $W_{37.5}Ta_{28.9}V_{17.8}Ti_{15.8}$, changed to $W_{31.5}Ta_{30.9}V_{21.4}Ti_{14.3}$ after the reaction. (These EDS maps were contrast-adjusted to enhance the visibility of elemental distributions. More accurate compositions for each region, measured by point EDS analysis, are presented in Table 1.) Following the reaction, the W content decreased by ≈6 at%, while the V content increased by ≈4 at%, with minor compensatory adjustments observed in the concentrations of the remaining elements. Despite these changes, the overall compositional variation remains limited, and the measured properties of the RHEA dendrites in the precursor are expected to be preserved in the final composite.

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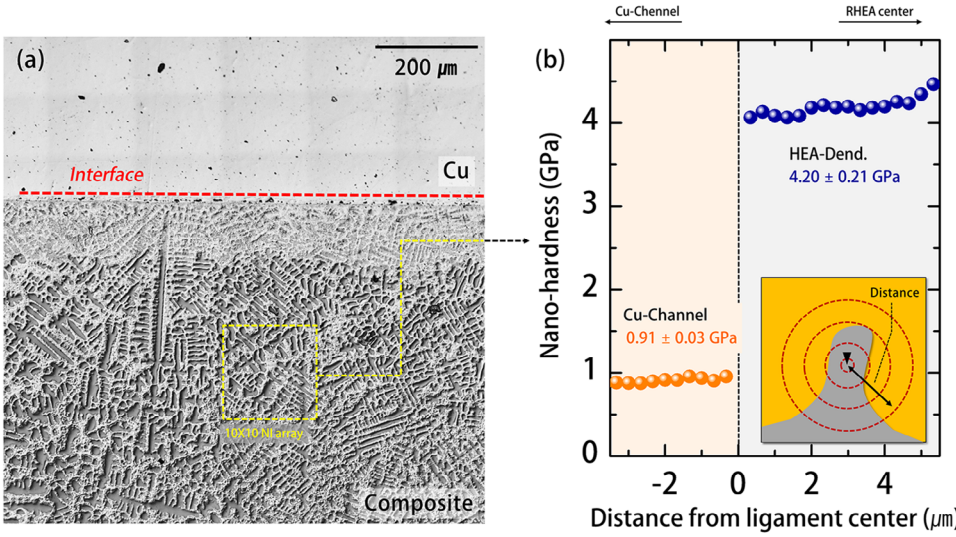


Figure 5. HEA-Cu composite with a diffused interface fabricated via the LMD process. a) Overall image of the HEA-Cu composite captured using OM after the LMD process. b) Nanoindentation measurements of the region, with nano hardness plotted as a function of the distance from the HEA ligament center.

Slight reductions in W and Ti content in the dendritic regions indicate minor Ti extraction during the LMD process, attributed to its reactivity with Cu. However, the overall RHEA composition in the dendritic regions remained stable within ±5 at.%, confirming that atomic exchange was localized primarily in the ID regions. Although trace amounts of RHEA elements (<0.35 at.%) were detected in the ID regions, they were negligible impurities. The Cu melt contained less than 0.1 at.% Ti, which did not significantly affect its κ, as impurities at such levels have minimal impact.[36]

The composite structure formed via the LMD process was distributed throughout the material rather than limited to the surface. **Figure 5**a demonstrates the Cu melt's penetration extended millimeters into the material, underscoring the feasibility of scaling this heat-conductive composite for larger components. Even at the macroscopic scale, it was clearly observed that Cu selectively replaced only the interdendritic regions while the overall dendritic architecture remained intact. This preservation of structural anisotropy is expected to play a critical role in enhancing the thermal transport properties of the composite. As presented in Figure S4 (Supporting Information), the Cu phase arrangement in the RHEA–Cu composite—fabricated from a dendritic-structured precursor—was compared to that of a Ta–Cu composite derived from a disordered solid solution precursor. The RHEA–Cu composite exhibited a arrangement-coherence value of 0.274 (ranging from 0 to 1, with 1 representing perfect alignment), markedly higher than that of the randomly distributed composite (0.054), indicating a significantly greater degree of directional alignment.

Figure 5b plots the hardness (H), measured by nanoindentation, as a function of the distance from the nearest RHEA ligament. The results distinctly show that hardness varies depending on the phase. In particular, the hardness of the Cu-channel region is nearly identical to the known hardness of pure Cu, ≈1 GPa.[37a] This consistency suggests that the LMD process successfully yielded a high-quality composite structure with inter-ligament layers of nearly pure Cu composition, as designed.

For effective heat conduction, examining the interface between the two phases is critical. **Figure 6** presents the microstructure and EDS mapping results analyzed using TEM to closely investigate the interface of the fabricated composite. Consistent with the SEM analysis, the TEM analysis confirms that the interface of this alloy is distinctly separated into RHEA and Cu regions, as shown in Figure 6(a-1). Figure 6b–e further demonstrate that W, Ta, V, and even Ti primarily reside in the dendritic ligaments of the RHEA.

Notably, TEM analysis identifies two additional microstructural features not observed in the SEM images from Figure 4. The first feature is a W-rich region. As shown in the figure, the interlayer ≈300 nm thick was observed at all interfaces between the RHEA dendrites and the surrounding Cu. (A more detailed SEM images that we did the TEM-sampling of the interface has been provided in Figure S5, Supporting Information). As confirmed by TEM-EDS analysis, this interlayer predominantly comprises W, which correlates with the observed decrease in the average W composition in the dendritic region after the reaction. The second feature is the formation of Ti-rich precipitates within the HEA interface. This phenomenon is attributed to the recrystallization of Ti, dissolved at the interface during the reaction, into an HCP phase during the cooling process. The XRD trace in Figure 4 also indicates the presence of a small fraction of HCP phases following the LMD process, corroborating the formation of these HCP-Ti precipitates. These Ti precipitates reduce the amount of Ti dissolved in the dendritic region and proportionally increase the relative concentrations of W and Ta, which are higher-conductivity elements, potentially enhancing the κ of the composite.

The formation process of this unique structure is illustrated step-by-step in the schematic diagram in Figure 6f–k. (For simplicity, the RHEA elements with positive ΔH_{mix} are represented by W, while those exchanged during the process are represented

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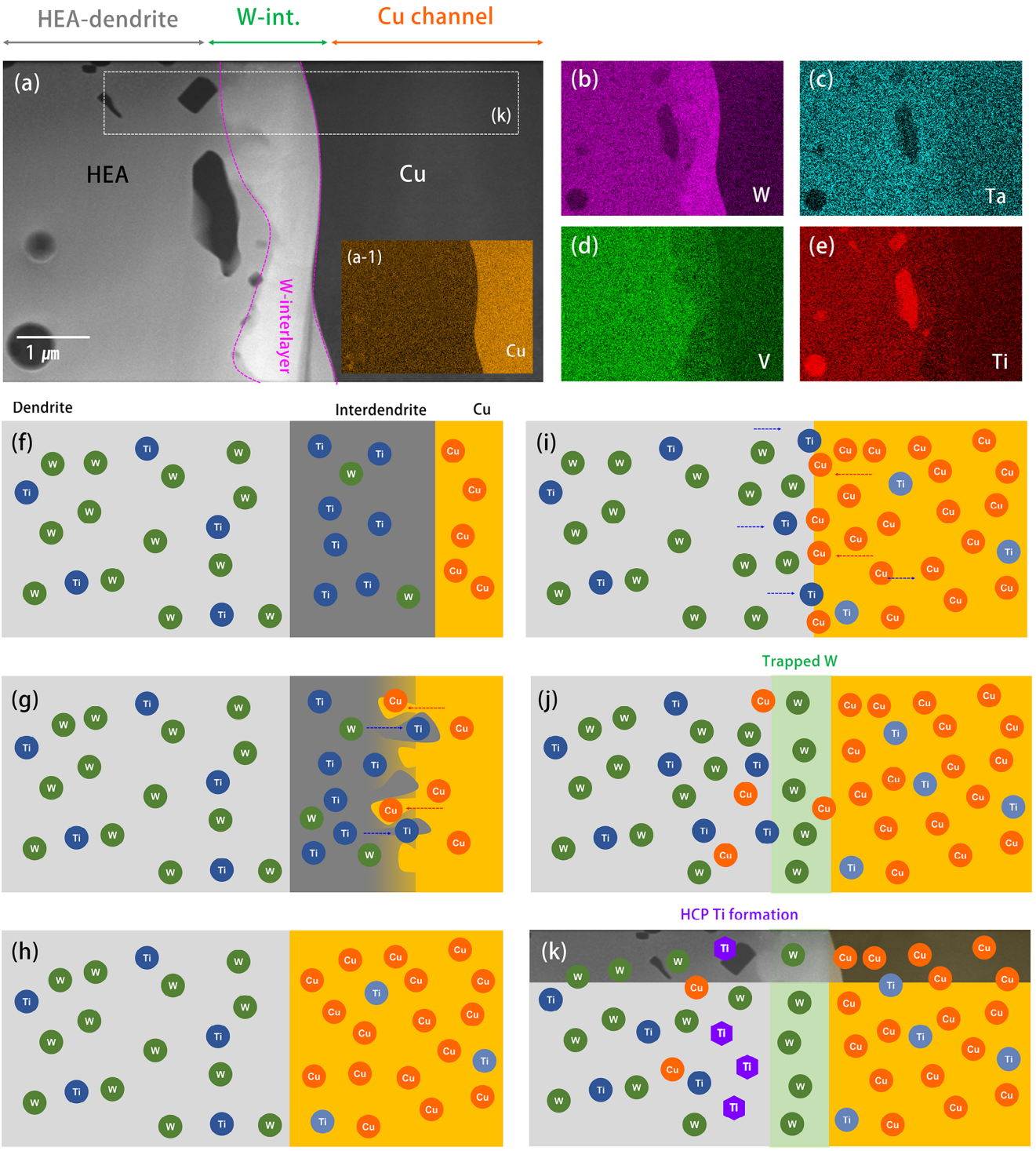


Figure 6. Microstructure of the interface between HEA and Cu channel after the LMD process. a) TEM image of the HEA-Cu interface and corresponding EDS maps: b) W, c) Ta, d) V, e) Ti. f–k) Mechanism of interface microstructure formation in the HEA-Cu composite.

by Ti.) Initially, when the RHEA with a D-ID structure reacts with the high-temperature Cu melt, Cu preferentially reacts with the Ti-rich ID regions (f). Although the Ti in the D regions may also react with Cu, the ID regions contain more than twice as much Ti and exhibit a lower melting point, making it thermodynamically and kinetically more favorable for the reaction to proceed primarily in the ID regions. As a result, the early stage of the reaction predominantly occurs in the ID regions. During this process, Ti, which is miscible with Cu, rapidly exchanges positions at the interface and dissolves into the melt, while Cu atoms

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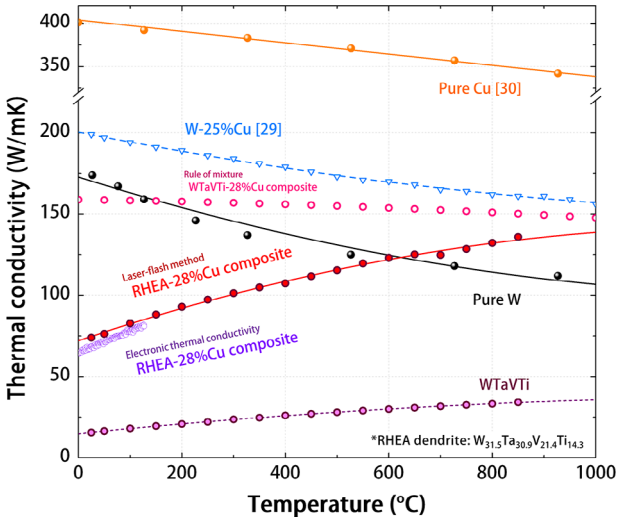


Figure 7. Temperature-dependent thermal conductivity of various alloys. This figure includes the thermal conductivity of pure Cu and W, the WTaVTi alloy, its composite with Cu, as well as previously reported W-Cu composites.

occupy the original positions of Ti in the ID regions (g). After sufficient reaction time, all Ti in the ID regions dissolves into the Cu melt, leaving Cu to replace the ID regions. Consequently, the D regions become exposed to the Cu melt (h). With ≈15% Ti still present in the D regions, the remaining Ti also tries to participate in position-exchange reactions with Cu (i). As a result, the interface near the Cu channel of the RHEA becomes enriched with elements exhibiting positive ΔHmix with Cu. Due to the negligible solubility of these elements in Cu, further position-exchange reactions cease. Among these, W, which has the highest positive ΔH_{mix} with Cu, remains at the interface, forming a trapped W-rich interlayer between the two phases (j). During the subsequent cooling process, residual Ti solidifies slowly and recrystallizes behind the W-rich interlayer into HCP-phase precipitates (k).

2.4. Thermal Conductivity of the Fabricated RHEA-Cu Composite

Figure 7 presents the κ of various alloy materials measured over a range of temperatures. As shown, pure Cu and W exhibit exceptionally high κ at room temperature, ≈398 and 173 W m^{-1}·K^{-1}, respectively. However, consistent with most metals, their κ decreases as temperature increases. Within the temperature range analyzed, pure Cu and W show average declines with slopes of ≈−0.047 and −0.071 W m^{-1}·K^{-2}, respectively. This reduction is attributed to increased lattice vibrations at higher temperatures, which interfere with the mobility of conduction electrons.

In comparison, W-Cu composite alloys demonstrate intermediate κ values. For instance, a composite containing ≈25 wt.% Cu exhibits a κ of ≈200 W m^{-1}·K^{-1} at room temperature. Similar to pure metals, the κ of this composite decreases with temperature at a slope of −0.044 W m^{-1}·K^{-2}, reaching ≈155 W m^{-1}·K^{-1} at 1000 °C. This value corresponds to 80% of the κ predicted by the Rule of Mixtures (ROM), highlighting minimal heat loss caused by interfaces or microstructural features. These characteristics qualify the material as an excellent candidate for thermal management applications.

The WTaVTi alloy, which forms the ligaments of the LMD-fabricated composite, has a significantly lower κ, measuring ≈17 W m^{-1}·K^{-1} at room temperature, only ≈4% of pure Cu's κ. Unlike conventional metals, however, the κ of WTaVTi increases with temperature, displaying a positive slope of +0.022 W m^{-1}·K^{-2} and reaching a maximum value of ≈35 W m^{-1}·K^{-1} at temperatures above 800 °C. A similar temperature-dependent increase in t κ has also been reported in other HEA systems. This behavior is known to be attributed to the inherently highly distorted lattice structure of HEAs, where the increase in lattice-mediated heat transport with temperature outweighs the decrease in electronic κ.[37b] In contrast, the introduction of Cu in the RHEA by the LMD process enhances the κ drastically. The resulting RHEA-Cu composite achieves a κ of ≈75 W m^{-1}·K^{-1} at room temperature, 4.4 times higher than WTaVTi alone. However, this value corresponds to only ≈40% of the ROM-predicted κ (pink dots in Figure 7), suggesting greater heat loss compared to W-Cu composites. Despite this, the RHEA-Cu composite's κ increases more rapidly with temperature, with a positive slope of +0.075 W m^{-1}·K^{-2}, reaching ≈140 W m^{-1}·K^{-1} at 800 °C. At this temperature, the composite achieves over 85% of the ROM-predicted κ, indicating improved efficiency at elevated temperatures. In contrast, pure W exhibits a steep decrease in κ with temperature due to increased scattering of free electrons—the dominant heat carriers—at higher temperatures. Because of this, the κ of the LMD-fabricated RHEA–Cu composite surpasses that of pure W (≈120 W m^{-1}·K^{-1}) at ≈650 °C, and is expected to outperform conventional W–Cu composites at temperatures exceeding 1000 °C. Also, this trend suggests that the composite could outperform conventional W-Cu composites in κ at temperatures exceeding 1000 °C. These findings underscore the composite's potential for high-performance thermal management applications in extreme environments, making it a promising material for next-generation thermal dissipation systems.

3. Discussion

3.1. Enhancement of Irradiation Resistance of the RHEAs

The effects of high-dose materials α-ion irradiation, as well as transmutation effects, can be studied using ion beam irradiation. Accelerated α-ions can create defects in the materials, which evolve through diffusion and recombination into gas clusters or voids, ultimately causing volume swelling largely driven by the Gas content. Once Bubbles connect, they can lead to micro-scale cracks on the surface, resulting in blistering as observed by many studies.[32,34,38,39] These macroscopic structural changes, including swelling and blistering, are critical criteria for evaluating materials intended for harsh irradiation environments[34] as they are found in fusion environments. However, it is important to note that the Helium swelling due to high-dose helium exposure is not to be mistaken for void swelling which is a vacancy and interstitial driven phenomena. In this work we only consider the Helium effects directly.

As shown in Figure 3b, both W and the WTaVTi alloy exhibit minimal swelling up to a fluence of ≈4 × 10^{17} atoms cm^{-2} (≈29% of He concentration), after which swelling increases rapidly, with

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blistering observed at ≈8 × 10^{17} atoms cm^{-2} for W and ≈8.5 × 10^{17} atoms cm^{-2} (≈60% of He concentration) for WTaVTi. This rapid void growth after a certain dose suggests that initially isolated irradiation-induced defects evolve into macro-scale cavities.[40] Compared to W, the WTaVTi alloy exhibits over 30% less swelling, highlighting its suitability for more severe irradiation conditions. This superior resistance can be partly attributed to the RHEA's high strength.

Fan et al. proposed that the internal pressure (**p**) within blisters can be used to predict the stress at which cracks form on the blister surface. At ambient temperatures, vacancy diffusion within the blister region is effectively negligible due to the high activation energy required for vacancy migration in W-based refractory alloys; thus, understanding the internal pressure becomes crucial in evaluating crack initiation. Equation (1) calculates the internal pressure of overpressed bubbles, considering the bubble spacing (**d**), radius of a bubble (**r**), surface tension (γ), and average tensile stress (**T**) across a bubble plane:[41]

$$p = \frac{2\gamma}{r} + \frac{T\left(d^2 - \pi r^2\right)}{r^2} \tag{1}$$

To simplify the calculations, several assumptions were made. It was assumed that the WTaVTi alloy has similar r (≈1 nm) and d (≈2 nm) values as W. Although precise values of γ for RHEAs are unattainable, the surface tensions of refractory elements such as W, Ta, V, and Ti have been reported. Using the ROM, which provides accurate γ estimates for high-alloy binary systems, the γ for WTaVTi was approximated as the average of its constituent elements. The reported γ values for W, Ta, V, and Ti are ≈2.83, 2.68, 2.15, and 1.61 N m^{-1}, respectively, resulting in an estimated γ of ≈2.32 N m^{-1} for WTaVTi.[42–45a] The tensile stress (**T**) was replaced with the compressive YS from Figure 3a, as theoretical YS values are assumed to be equivalent under tensile and compressive stress. For W and WTaVTi, the YS values used were 563 and 1467 MPa, respectively. In Equation (1), the left-hand term represents the maximum pressure sustainable due to surface tension, while the right-hand term reflects the material's resistance to cracking based on its strength. Comparing these terms elucidates their respective contributions to blister resistance.

By comparing the left-hand term ($\frac{2\gamma}{r}$) and right-hand term ($\frac{T(d^2-\pi r^2)}{r^2}$), the contribution of each factor can be assessed. The calculated total **p** for each material are 6.14 and 5.89 GPa, showing a negligible difference of ≈4%. However, for W, which has a relatively high γ, the contributions from the left- and right-hand sides were 5.66 and 0.48 GPa, respectively. In contrast, for the WTaVTi alloy, which exhibits higher material strength, the corresponding values were 4.73 and 1.26 GPa, with the right-hand value being approximately three times larger than that of W. This comparison highlights that superior irradiation resistance can be achieved by combining high surface energy, as seen in W, with enhancement in strength by alloying. This finding underscores the potential for advancements in material performance not merely by relying on pure W but through the further optimization of RHEA compositions, enabling the development of improved materials for extreme environments.

3.2. Manipulation of Microstructure in RHEA-Cu Composite by Process Optimization

During the LMD process, the content of sacrificial materials in the precursor is predetermined, which typically means that the final phase fraction remains constant regardless of the process conditions, such as the LMD temperature and time. However, the thickness and morphology are known to vary based on the reaction parameters.[15] In this study, it was assumed that Cu would only react with the Ti-rich ID regions, suggesting that neither the thickness nor the phase fraction of the ligaments would be significantly affected by the reaction conditions. Nonetheless, the reaction performed at 1200 °C for an extended duration of 96 h revealed some deviations from this assumption. As shown in Table 1, the initial volume ratio of the D-ID regions (≈71:29) shifted to a post-reaction volume ratio of ≈64:37 between the RHEA ligament and Cu channel. This reduction in the RHEA dendrite volume indicates that some unintended dissolution reactions occurred within the RHEA ligaments due to residual Ti. In fact, around the dendrite regions in Figure 4b, there exist isolated features on the order of several hundred nanometers that exhibit a contrast similar to that of the dendrites. These are interpreted as additional RHEA regions formed through reactions between residual Ti within the dendrites and the surrounding Cu. Therefore, if the reaction time is further extended, it is possible that even the Ti remaining in the current RHEA ligaments may eventually react with Cu, potentially leading to the formation of a dual-ligament structure. This observation highlights the need for further studies aimed at optimizing the reaction conditions to control such microstructural evolution.

The compositional changes are evident in the slight reduction in Ti content within the dendrite regions, as shown in Table 1. This suggests that, after the complete reaction between Cu and the ID regions, the Cu melt continued to react with residual Ti in the dendrites over the long reaction period. ≈15% of the Ti, a negative ΔH_{mix} element, remained in the dendrite regions, which could potentially engage in further reactions. Adjusting the reaction temperature or extending the reaction time could enable additional reactions between Cu and the dendrite's residual Ti, allowing for secondary Cu channels within the RHEA ligaments to form.

Meanwhile, despite the relatively low Ti content (≈36%) in the ID regions ($W_{9.9}Ta_{17.7}V_{35.8}Ti_{36.6}$), it was observed that all alloying elements in this region could be completely dissolved into the Cu melt through their interaction with Cu. Even W, which has a very low solubility in liquid Cu (≈0.5–1 at.%), could dissolve into the melt under these conditions. Conversely, the D regions ($W_{37.5}Ta_{28.9}V_{17.8}Ti_{15.8}$) exhibited much slower reaction rates, with the reaction limited to the surface even at 1200 °C. Previous studies have shown that even small fractions of sacrificial elements (<20 at.%) can enable sufficiently fast reactions.[17] The reaction rate during the LMD process is primarily governed by interatomic diffusion between elements, which, in turn, depends on the melting point differences between the phases.

Therefore, it can be understood that various factors beyond mere thermodynamic relationships influence the kinetics of the LMD reaction. In particular, the melting point (T_m) of a material serves as a critical factor in determining diffusion kinetics and can significantly affect the rate of the LMD process. The T_m

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of the D and ID regions, calculated via CALPHAD simulations, were ≈2623 and ≈2117 K, respectively. Given the reaction temperature of 1200 °C in the molten Cu bath, these correspond to ≈0.55Tm for the D regions and ≈0.70T_m for the ID regions. For faster reactions, the sacrificial material should ideally have a melting point corresponding to ≈0.7T_m relative to the molten metal bath. For example, lowering the T_m could restrict reactions in the D regions, while adding low-melting-point alloying elements or increasing the Cu melt temperature could promote additional reactions in the D regions. These adjustments provide a pathway for tailoring the microstructure and phase distribution within the final RHEA-Cu composite, enabling process optimization to achieve the desired material properties.

Meanwhile, in this study, the reaction temperature was limited to 1200 °C to ensure the stability of the quartz tube during the prolonged reaction process. Additionally, since the LMD reaction proceeds from the surface and achieving a homogeneous microstructure throughout the entire material is critically important for composite performance, the reaction time was deliberately extended to 96 h—significantly longer than in typical LMD-related studies.[15–17,45b] As a result, a highly uniform microstructure was successfully achieved on the mm-scale, as shown in Figure 5a. Nonetheless, to move toward practical implementation of this material, faster diffusion kinetics could be induced by increasing the reaction temperature. In fact, a previous study by Ian McCue et al. demonstrated that, for a similar Ta–Ti alloy system reacted in a Cu melt, the reaction depth was ≈245 μm at 1160 °C, whereas increasing the temperature to 1360 °C led to a depth of 353 μm—corresponding to more than a 40% enhancement in reaction rate. These findings suggest that further optimization studies could open pathways for scaling up the process and significantly reducing processing time, ultimately enabling more efficient fabrication of such composites.

3.3. Mechanism of the Formation of Self-Constructed W-Interlayer

The spontaneously formed W-interlayer region exhibits a higher concentration of W compared to other dendrite regions, which contain a more balanced ratio of W and Ta. This selective formation of the W-rich interlayer is attributed to differences in the diffusion rates of the constituent elements. Generally, the mobility of elements within an alloy is governed by an Arrhenius-type equation, with the activation energy required for diffusion to a vacancy site playing a significant role in determining the diffusion rate. According to simulated self-diffusivity values by Zhou et al., at a temperature of 1500 K—similar to the reaction temperature of 1200 °C in this study—Ta exhibits ≈8×10^{14} cm^2 s^{-1}, whereas W shows a significantly slower diffusivity of less than 1×10^{18} cm^2 s^{-1}, a difference of over four orders of magnitude.[46] These values represent self-diffusion in pure Ta and W; within the distorted lattice structure of a RHEA, Ta, with its lower melting point and smaller atomic size, would be expected to diffuse even faster. During the reaction at the Cu melt interface, as Ti diffuses out of the ID region and into the Cu melt, the surface of the ligament becomes temporarily enriched with other constituting elements. Upon cooling, the slower diffusion rate of W prevents it from homogeneously redistributing within the ligament. Instead, W becomes concentrated near the surface, spontaneously forming the W-rich interlayer. This trapping effect of W can be understood as a direct consequence of its limited diffusivity.

The formation of the W-enriched region also facilitates the precipitation of HCP-structured Ti-rich phases. Figure 6k illustrates the final structure of the interface, highlighting these precipitates. Notably, Ti remaining near the W-interlayer during the cooling process precipitates as an HCP phase, a thermodynamically stable structure for Ti and V at temperatures below 1200 K (as indicated by the blue region in Figure 2b). The EDS maps confirm that these precipitates are Ti-rich with small amounts of V, consistent with the stable HCP phase at these temperatures. (Figure S6a, Supporting Information shows the binary Ti-V phase diagram.) The potential locations for these precipitates are conceptually overlaid in Figure 6a, marked by dashed lines. This aligns with observations of conventional Ti alloys, such as Ti-64, which readily form HCP phases during slow cooling processes like furnace cooling. The over-saturated Ti at the Cu interface during solidification likely transformed into the HCP phase. Even if the LMD process temperature was as high as 1473 K, the gradual cooling within the furnace would still enable the formation of HCP precipitates in the Ti-rich interface regions, as supported by the CCT diagram for Ti-64 alloys shown in Figure S6b (Supporting Information). However, the volume fraction of these precipitates was found to be less than 1 vol.% and was confined to the surface of the RHEA ligaments adjacent to Cu. The Cu channels, which play a dominant role in thermal transport, showed minimal evidence of these precipitates. Therefore, it is expected that their impact on the overall $\boldsymbol{\kappa}$ of the composite is negligible. This localized formation of HCP phases represents a minor microstructural feature that does not detract from the thermal efficiency of the material.

3.4. Thermal Conductivity and Interfacial Resistance Analysis of RHEA-Cu and W-Cu Composites

The κ of the composite is influenced by the κ values of its constituent phases and the energy loss at the interfaces during heat transfer. The RHEA-Cu composite developed in this study shows relatively low κ at room temperature, as depicted in Figure 7. This suggests that the RHEA-Cu has significant interfacial energy losses in heat transfer at room temperature. However, the κ increases steeply with temperature (+0.075 $W\ m^{-1}{\cdot}K^{-2}$ slope) and surpasses that of pure W above 650 °C, indicating a substantial reduction in interfacial energy losses with increasing temperature.

Interfacial resistance can be explained by theories such as the Acoustic Mismatch Model (AMM) and the Diffuse Mismatch Model (DMM).[47] The AMM attributes resistance to acoustic reflections due to crystallographic differences but is less suitable for metallic systems where electron-driven heat transfer dominates. Instead, the DMM, which considers electron scattering at interfaces, provides a more accurate explanation in this case. Using the Wiedemann-Franz law ($\boldsymbol{\kappa = \sigma LT}$) and the measured electrical resistivity (via four-point probe), the electronic contribution to $\boldsymbol{\kappa}$ was calculated, and shown as pink dots in Figure 7. The resulting values closely match those obtained through laser flash analysis (LFA), differing by less than 5%, confirming that electron conduction predominantly governs heat transfer in this MMC.

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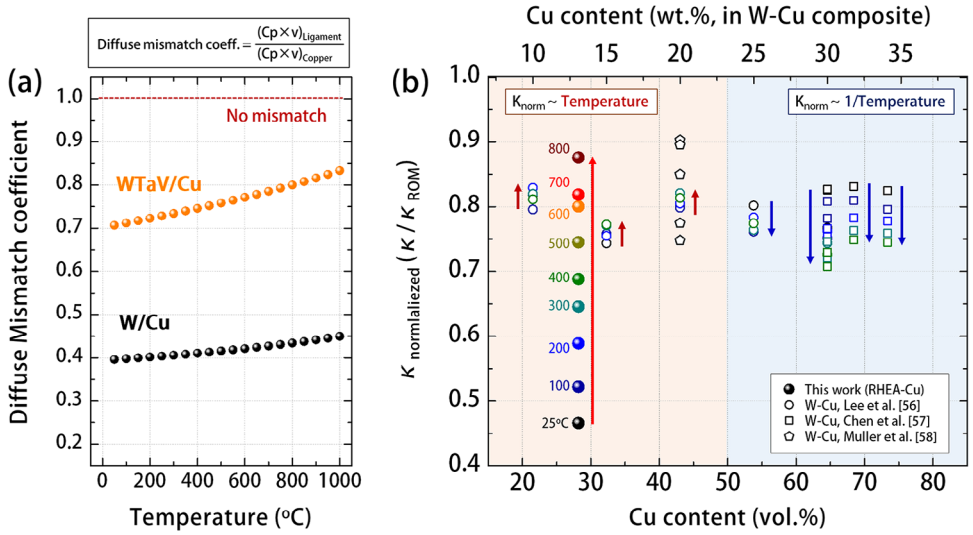


Figure 8. Thermal conductivity and interfacial resistance changes in composite materials based on the DMM model. a) Temperature-dependent mismatch between WTaV or W and Cu as predicted by the DMM model. b) Normalized thermal conductivity (TC) of HEA-Cu and W-Cu composites as a function of Cu fraction, along with the temperature-dependent thermal conductivity for each fraction.

According to DMM, the interfacial thermal efficiency can be quantified using the Diffuse Mismatch Coefficient (DMC), calculated as the ratio of the specific heat (**Cp**) and sound velocity (**v**) of the ligament phase to that of the Cu channel as Equation (2) below.

$$Diffuse\ mismatch\ coefficient = \frac{(C_p \times v)_{ligament}}{(C_p \times v)_{Cu\ channel}} \quad (2)$$

Figure 8a illustrates the calculated DMC values for W-Cu and RHEA-Cu composites. The input values for **Cp** and **v** were obtained from the literature, and RHEA ligament (WTaV) properties were estimated using ROM, assuming complete dissolution of Ti into the Cu melt.[48–59a] (The calculated values for each variable that determines the DMC are detailed by temperature in Figure S7, Supporting Information). At room temperature, the DMC for WTaV-Cu was ≈0.7, significantly higher than that of W-Cu (≈0.39). The DMC for WTaV-Cu increased with temperature, reaching ≈0.84 at 1000 °C, while the W-Cu DMC remained relatively low (≈0.45). This indicates that heat loss at the RHEA-Cu interface diminishes with increasing temperature, improving thermal transfer efficiency. In contrast, W-Cu composites exhibit less temperature-dependent interfacial resistance, leading to smaller changes in κ.

It should be noted, however, that the actual interface in the RHEA–Cu composite may include a W-rich interlayer rather than a sharply defined WTaV–Cu boundary. However, such interlayers are typically thin and formed through gradual dealloying, resulting in compositionally graded structures. These compositionally graded interfacial structures have been shown to facilitate phonon transmission by smoothing abrupt property discontinuities, thereby mitigating acoustic impedance mismatch. For instance, Ding et al. demonstrated that a W–Cu graded layer fabricated via selective laser melting enhanced both thermal stress tolerance and phonon transport continuity across the interface.[59b] Similarly, Zhang et al. reported that diffusion-bonded W/Cu interfaces with Ni interlayers formed alloyed interphases that improved thermal coupling by reducing interfacial phonon scattering.[59c] Moreover, DMM calculations are based on bulk transport properties rather than atomic-scale ordering, and therefore remain valid in the presence of such graded interfaces. As such, we consider our DMM-based estimation to be physically reasonable and appropriate for this system.

Figure 8b compares the normalized thermal conductivities (κ_{norm}) of various composites, where κ_{norm} represents the κ normalized by the theoretical maximum (ROM). At room temperature, RHEA-Cu composites exhibited a low κ_{norm} (≈0.47), dominated by the less conductive RHEA ligament. However, at 800 °C, κ_{norm} increased to ≈0.88, reflecting reduced interfacial losses and enhanced Cu channel contribution. Conversely, κ_{norm} for W-Cu composites showed minimal change with temperature, consistent with the relatively invariant DMC values.

The trends in κ_{norm} also varied with Cu fraction. Typically, the κ of a composite exhibits a sigmoidal curve as a function of the volume fraction of the secondary phase.[45] At low fractions of the secondary phase, the thermal energy predominantly travels through the matrix material, as the secondary phase lacks sufficient connectivity. Thus, the overall κ is determined by the matrix's properties, with negligible contributions from the secondary phase. As the fraction of the secondary phase increases and both phases achieve sufficient connectivity, the κ aligns more closely with the ROM, accounting for energy losses due to carrier scattering at the interfaces. For W-Cu composites with <50% Cu, κ_{norm} slightly increased with temperature due to enhanced contributions from Cu. For composites with >50% Cu, κ_{norm} decreased significantly (≈15–20%) at 400 °C compared to room temperature, as heat transfer became dominated by W's lower κ. This highlights the matrix's dominant role in determining overall κ.

While DMM cannot fully explain all κ phenomena, the trends observed align well with the diffuse mismatch theory. The RHEA-Cu composite, which initially showed lower κ than W-Cu at room temperature, exhibited a dramatic increase with

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temperature, surpassing W-Cu composites with similar Cu fractions at ≈500 °C. Additionally, the composite's κ exceeded 120 W $m^{-1}\cdot K^{-1}$ at 600 °C, outperforming pure W. This temperature range aligns with the operating conditions of molten salt reactors, demonstrating the potential of RHEA-Cu composites for high-performance thermal management applications in extreme environments.

4. Conclusion

This study successfully developed a novel RHEA-Cu composite using the LMD process, marking a pioneering attempt in material science. The composite exhibits exceptional mechanical properties and κ under high-temperature conditions, presenting an innovative solution that overcomes the limitations of existing materials. This advancement is highly significant as it marks the development of an exceptional nuclear material that combines remarkable mechanical strength, superior irradiation resistance, and excellent thermal conductivity at elevated temperatures. The primary findings of this research are summarized as follows:

1) The WTaVTi alloy also exhibited outstanding irradiation resistance, with 30% lower swelling under α-ion irradiation compared to pure W, a benchmark material for irradiation-resistant metals. Remarkably, even at ≈57 at.% of He implantation, no swelling or blister-induced surface cracks were observed, indicating its potential for extensive use in nuclear applications, where materials are exposed to severe irradiation damage.
2) The composite structure optimized heat conduction pathways through a networked microstructure between the RHEA and Cu phases. Unlike traditional LMD processes, this study employed as-cast precursors with a highly directional D-ID structure. This approach deliberately replaced ID regions with Cu to form a directional composite structure, significantly enhancing κ and the bonding performance between plasma-facing materials (PFM) and water-cooled Cu pipes.
3) The LMD process was performed over 96 h at 1200 °C, enabling the reaction in mm-scale. This achievement contrasts sharply with conventional LMD processes, which are typically limited to surface-scale reactions. Importantly, the potential of this LMD approach is not limited to a specific system. In particular, HEAs, with their vast compositional diversity, offer far more variation than conventional alloy systems. This opens up opportunities to fabricate entirely new classes of composites or porous materials that were previously unattainable. The success of this long-duration, large-volume fabrication represents a critical breakthrough toward industrial-scale production and commercialization.
4) The process also led to the spontaneous formation of a unique W-rich interlayer at the RHEA ligament surface. Unlike commercial W-Cu materials, which often require additional steps to incorporate interlayers for improved interfacial bonding, the W-rich interlayer in this study formed autonomously. This feature not only simplifies the process but also ensures excellent interfacial bonding between the two phases, minimizing defects such as voids and reducing κ losses.
5) While the RHEA-Cu composite exhibited lower κ at room temperature compared to W-Cu composites with equivalent Cu fractions, it showed a steep increase with temperature, reaching ≈120 W $m^{-1}\cdot K^{-1}$ at ≈650 °C, surpassing pure W. This temperature coincides with the operating range of typical molten salt reactors (600–700 °C), highlighting the composite's suitability for nuclear applications. The temperature-dependent increase in κ can be attributed to the rising DMC between RHEA and Cu, which improved from 0.47 at room temperature to 0.88 at 800 °C, making the composite highly effective at elevated temperatures.

The RHEA-Cu composite developed in this study represents a groundbreaking advancement in high-temperature materials, excelling in mechanical strength, irradiation resistance, and κ. This composite is poised to become a next-generation high-temperature material, with potential applications in nuclear and aerospace industries and beyond. Its unique properties position it as a crucial solution for extreme environments, as well as industries where high-temperature performance and efficient thermal management are essential.

5. Experimental Section

Alloy Fabrications: RHEAs, W, WTa, WTaVi, and WTaVTi, were synthesized using an arc-melter. High-purity elements (≥99.9%), W, Ta, V, Ti, and Cu were melted under a Ti-gettered argon atmosphere. To ensure homogeneity, the ingots were melted at least five times, followed by solidification and cooling on a water-cooled copper (Cu) hearth. The cast alloy ingots were machined into suitable sizes for each experiment using a precision high-speed saw with a diamond-metal bonded wafer blade or an electrical discharge machine (EDM). The surfaces of the machined samples were polished with sandpaper up to P2000 grade to minimize the influence of surface roughness during the LMD reaction.

Liquid Metal Dealloying (LMD) Process: For the LMD process, the precursor alloy (prepared in a coin shape, 1 mm thick, 4 mm diameter, ≈1 g weight) and 100 g of pure Cu (99.999%) were encapsulated in a silica tube under vacuum. The tube was evacuated using a rotary pump. The precursor contained ≈0.1 g of Ti, and any dissolution into the Cu melt was expected to have a negligible effect on Cu purity. To maintain Cu in a liquid state, the tubes were placed in a muffle furnace at 1200 °C for 96 h. After the reaction, the silica tube was quenched in room-temperature water to cool the material as quickly as possible. Although cracks were observed on the silica tube's surface during cooling, the tube remained robust, preventing direct cooling of the material by the water. Therefore, the precise cooling rate could not be determined. After the reaction, the RHEA-Cu composite was recovered by grinding away the surrounding Cu parts.

Mechanical Testing: Compressive tests were conducted on a tensile testing machine (Instron 5967, Instron Inc., USA) at a strain rate of 10^{-3} s^{-1}. The tests covered temperatures ranging from 77 to 1073 K and were terminated when compressive strain exceeded 30%. Rectangular specimens with dimensions of 2 mm × 2 mm × 4 mm were used, and results were averaged over at least three tests.

The nano-hardness of the materials was measured via nanoindentation (TI 950 Triboindenter, Bruker, USA) using a Berkovich indenter tip. Indentation was performed with a trapezoidal load function, applying a maximum load of 10 mN for 5 s, holding at maximum load for 2 s, and unloading over 5 s. Hardness was calculated from the unloading curve. Indentations were arranged in a 10 × 10 array with a 5 µm spacing between points, resulting in measurements from 100 points, which were plotted as a function of distance from the ligament center.

Microstructural Characterization: The crystal structure of the specimens was analyzed using an X-ray diffractometer (D2 Phaser, Bruker) with Cu Kα1 radiation (λ = 0.15406 nm). Scanning was performed at a speed of 15° per min, with an exposure time of 5 s per degree to ensure

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high-intensity readings. Microstructural observations were performed using a field-emission scanning electron microscope (FE-SEM; Mira3, Tescan, Czech Republic) equipped with an energy-dispersive spectrometer (EDS; xFlash 6130, Bruker, Germany). Further detailed analysis was conducted using a transmission electron microscope (TEM; JEM-2100F, JEOL), which was also equipped with an EDS system (Ultim Max TLE, Oxford Instruments, UK) and Aztec TEM EDS software (Oxford Instruments, UK). TEM samples were prepared using a focused ion beam (FIB; Helios NanoLab 600, FEI, USA). FIB lamellae were bulk-out from the interface between the RHEA dendrite and Cu and prepared to a size of ≈10 × 20 µm. These samples were thinned to ≈100 nm and cleaned on both sides at 5 kV for final preparation.

Thermal and Electrical Property Measurement: Thermal conductivity (κ) was determined by measuring thermal diffusivity (α) using the laser flash method (LFA 457, NETZSCH, Germany) over a temperature range of 298–1173 K in an argon atmosphere. Specific heat (Cp) was measured via differential scanning calorimetry (DSC 8500, PerkinElmer, USA) at a heating rate of 10 K min^{-1}. The density of the alloys was calculated using Archimedes' principle with an electronic balance (XS204, Mettler Toledo, Switzerland). Tridecane was used as a medium for precise measurements due to its higher viscosity and lower density (0.756 g cm^{-3}) compared to water. The final density values were averaged from five measurements, excluding the maximum and minimum. The resistance of alloy samples (10 × 12 × 0.3 mm) was measured using a physical property measurement system (PPMS-14, Quantum Design, USA).

Ion Implantation and Swelling Measurement: The top surface of the disks was mechanically polished to achieve a root-mean-square surface roughness of ≈1 nm, verified by atomic force microscopy (AFM) after the final step of mechanical polish (VibroMet, Beuhler, Germany). Helium ion implantation was performed on the polished surface using a focused helium ion beam (ORION Helium Ion Microscope, Zeiss, Germany) at room temperature. Selected areas (10 × 10 μm^2) were irradiated with a beam energy of 25 keV and beam currents of 100 pA, resulting in a beam spot size of several nanometers. Irradiation doses ranged from 5 × 10^{16} to 1 × 10^{18} ions cm^{-2}, with repeated implantation to reach the desired doses.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords

composite, high thermal conductivity, high entropy alloys, irradiation resistance, liquid metal dealloying

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